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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (AI&DS/AI&ML/CCE/CSE/ISE) (CBCS) Degree Examinations

December 2023

CS2001-1 – DATA STRUCTURES

(For AI&DS, AI&ML, CCE, CSE, ISE)

Duration: 3 Hours

Max. Marks: 100

Note:

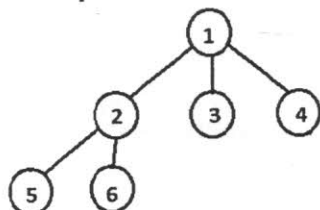
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

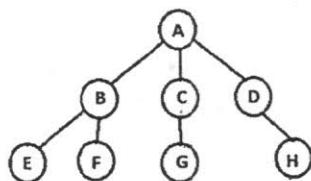
1. **How do you initialize an array in C?**
A) `int arr[3] = {1,2,3};` B) `int arr(3) = {1,2,3};`
C) `int arr[3] = {1,2,3};` D) `int arr(3) = (1,2,3);`
2. **What is the base data type of a pointer variable by which the memory would be allocated to it?**
A) `int` B) Depends upon the type of the variable to which it is pointing
C) No data type D) `unsigned int`
3. **Which of the following is correct way to declare multidimensional array in C?**
A) `int[] arr;` B) `int arr[][];`
C) `Int arr [[]];` D) `int[][] arr;`
4. **What is a sorted array?**
A) Array elements are in numerical order B) Array elements in alphabetical order
C) Elements of arrays in ascending or descending order D) All of the mentioned options
5. **What is the output of the ptr following C code?**
`char *ptr;`
`char mystring[] = "abcdefg";`
`ptr = myString;`
`ptr += 5;`
A) `fg` B) `efg`
C) `defg` D) `cdefg`
6. **What will be the output of the program?**
`void main() {`
`printf("%d %d", sizeof(int *), sizeof(int **));`
`}`
A) `2, 0` B) `0, 2`
C) `2, 2` D) `2, 4`
7. **Process of removing an element from stack is called _____**
A) Create B) Push
C) Evaluation D) Pop
8. **In a stack, if a user tries to remove an element from an empty stack it is called _____**
A) Underflow B) Empty collection
C) Overflow D) Garbage Collection
9. **Entries in a stack are "ordered". What is the meaning of this statement?**
A) collection of stacks is sort able B) Stack entries may be compared with the '<' operation
C) The entries are stored in a linked list D) There is a Sequential entry that is one by one

10. If the elements "A", "B", "C" and "D" are placed in a queue and are deleted one at a time, in what order will they be removed?
 A) ABCD
 B) DCBA
 C) DCAB
 D) ABDC
11. A queue follows _____
 A) FIFO (First In First Out) principle
 B) LIFO (Last In First Out) principle
 C) Ordered array
 D) Linear tree
12. A data structure in which elements can be inserted or deleted at/from both ends but not in the middle is?
 A) Queue
 B) Circular queue
 C) Dequeue
 D) Priority queue
13. Queues serve major role in _____
 A) Simulation of recursion
 B) Simulation of arbitrary linked list
 C) Simulation of limited resource allocation
 D) Simulation of heap sort
14. Non-leaf nodes are also called as
 A) Primary modes
 B) External modes
 C) Middle modes
 D) Internal modes
15. What is a hash function?
 A) A function has allocated memory to keys
 B) A function that computes the location of the key in the array
 C) A function that creates an array
 D) A function that computes the location of the values in the array
16. A linear collection of data elements where the linear node is given by means of pointer is called?
 A) linked list
 B) node list
 C) primitive list
 D) None of these
17. Which node is the parent node of Node 6?



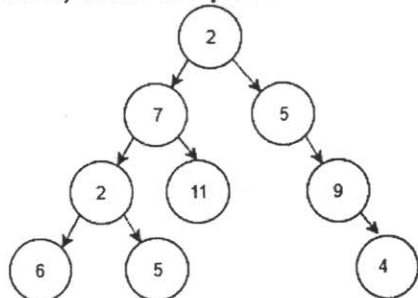
- A) 1
 B) 5
 C) 2
 D) 3

18. Which nodes are siblings of Node B of given ternary tree?



- A) E
 B) E and F
 C) F
 D) C and D

19. For the tree below, write the post-order traversal.



- A) 6, 2, 7, 2, 5, 11, 9, 5, 4
 B) 6, 5, 11, 2, 7, 5, 9, 4, 2
 C) 6, 5, 2, 11, 7, 4, 9, 5, 2
 D) 6, 2, 7, 2, 11, 5, 5, 9, 4
20. To obtain a prefix expression, which of the tree traversals is used?
 A) Level-order traversal
 B) Pre-order traversal
 C) Post-order traversal
 D) In-order traversal

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is Data Structure? Explain with example the primitive and non-primitive data structure with example.	06	L2	1	1
b)	What is Recursion? Explain with example.	04	L2	2	1
c)	Explain with the function and example any 2 primitive operations performed on the stack.	04	L1	2	1
d)	How array of Structures and Pointer to Structure is used in C language?	02	L2	1	1
2. a)	Write an algorithm to convert infix to postfix conversion. Explain with an example.	08	L2	2	1
b)	What is stack? List the characteristics of the stack.	04	L2	2	1
c)	Explain Tower of Hanoi problem. How it can be solved using the recursion?	04	L1	2	1
3. a)	Write a recursive program to find the factorial of a number. Also trace the program for n=4.	06	L2	2	1
b)	What is linear data structure? Explain the applications of the stack.	04	L1	1	1,2
c)	What is dynamic allocation? Explain the dynamic allocation of 'n' number of array and find the sum of the elements in an array.	06	L1	1	1,2
Unit – II					
4. a)	Explain the characteristics of queue. How is it implemented using the array? Explain with the example.	08	L2	3	1,2
b)	What is linked list? How is it implemented? Briefly explain the operations insert_after(p), delete_after (p) performed on the linked list.	08	L2	4	1,2
5. a)	Explain any 4 types of linked list with an example.	04	L1	4	1
b)	Implement Queue operations (using array) such as Insert, Delete and Display. Display appropriate messages on Queue overflow and Queue underflow conditions.	06	L2	3	1
c)	What are the advantages of circular linked list over linear list? Write the structures used to create a circular linked list and perform the basic operations.	06	L2	4	1,2
6. a)	Explain different types of queues with an example.	04	L1	3	1
b)	How linked list stack can be implemented? Write functions to implement the stack operations using linked list.	06	L2	4	1
c)	What are the advantages and disadvantages of doubly linked list? Write a C function to add a new node in doubly linked list considering all possible cases.	06	L2	4	1,2
Unit – III					
7. a)	What is a binary tree? Explain with an example. Write C functions to create a binary tree, and also to perform the following functions.	10	L2	5	1
	i) Preorder traversal ii) Inorder traversals iii) Post order traversals.	06	L2	5	1
b)	What is expression tree? Explain with an example.				
8. a)	With an example, explain hashing and also discuss different hashing functions used.	10	L2	5	1
b)	Write a short note on the following:				
	i) Threaded binary tree ii) Representation of the Graphs	06	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
